

ZLIKHA, KazakhstanWMO: **359690**Lat: **45.222N**Lon: **66.877E**Elev: **453**StdP: **14.46**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-8.3	-19.7	1.9	-12.8	-14.6	2.5	-6.7	39.9	21.1	35.3	9.7	5.0	50	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.2	101.5	66.7	98.7	65.9	95.9	65.1	69.6	93.6	68.2	92.2	66.8	90.8	8.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	86.7	76.4	59.8	78.1	76.3	57.2	70.9	76.3	33.9	93.2	32.7	92.3	31.6	90.8	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	31.0	22.9	19.6	DB	-18.5	106.6	7.9	3.2	-24.2	108.9	-28.8	110.7	-33.2	112.5	-38.9	114.8
(5)				WB	-19.5	73.8	8.3	3.8	-25.5	76.6	-30.3	78.8	-35.0	80.9	-41.0	83.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	16.7	20.9	38.5	55.2	67.1	76.7	79.6	77.2	64.4	49.4	34.0	21.6	
	DBStd	24.34	12.26	13.64	11.58	9.53	7.19	6.49	5.94	6.62	8.07	8.99	10.54	11.62	
	HDD50	3778	1033	814	380	57	1	0	0	0	4	123	486	880	
	HDD65	6805	1498	1234	822	313	62	4	0	4	108	486	929	1345	
	CDD50	3875	0	0	24	213	531	801	917	843	435	104	7	0	
	CDD65	1425	0	0	0	18	126	355	452	383	89	2	0	0	
	CDH74	20887	0	0	18	314	1897	5195	6714	5365	1296	88	0	0	
	CDH80	11251	0	0	3	86	777	2880	3927	3027	538	13	0	0	
Wind	WSAvg	7.4	8.5	8.2	8.7	8.1	7.9	7.0	6.7	7.1	6.7	6.1	6.3	7.9	
Precipitation	PrecAvg	9.9	1.2	1.2	1.1	1.0	0.9	0.6	0.5	0.2	0.2	0.6	1.1	1.2	
	PrecMax	17.7	2.5	2.7	2.1	2.4	2.9	2.3	1.6	0.9	0.6	2.2	2.3	2.9	
	PrecMin	5.4	0.4	0.2	0.2	0.2	0.2	0.1	0.0	0.0	0.0	0.1	0.2	0.4	
	PrecStd	2.9	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.4	0.3	0.2	0.5	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	53.8	76.1	88.4	96.3	103.5	106.7	104.1	97.5	82.0	65.6	47.2	
		MCWB	41.4	46.2	56.8	60.9	63.3	67.3	68.6	67.7	65.6	58.6	53.3	41.1	
	2%	DB	37.8	46.1	66.5	82.0	91.5	100.0	102.2	100.1	90.7	76.1	58.7	41.6	
		MCWB	35.3	40.6	51.5	58.0	62.3	66.3	66.9	66.2	62.1	55.1	49.1	37.8	
	5%	DB	34.6	40.8	60.5	77.2	88.0	96.8	98.9	97.2	86.4	71.2	53.1	38.3	
		MCWB	32.4	36.4	48.7	55.9	61.1	65.4	66.0	65.4	60.4	52.6	45.8	35.3	
	10%	DB	32.2	36.7	55.4	72.6	84.4	93.5	95.8	94.1	81.7	66.1	48.2	34.9	
		MCWB	30.3	33.1	46.0	54.4	60.1	64.5	65.4	64.4	58.3	50.3	42.6	32.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	47.0	57.0	63.8	68.3	71.7	71.3	71.6	66.6	59.4	55.4	42.3
			MCDB	45.3	53.4	72.8	80.0	84.6	92.5	94.6	93.3	93.6	77.9	61.7	45.7
2%		WB	35.6	41.0	53.1	60.5	65.4	69.2	69.4	68.8	63.3	56.5	49.9	38.3	
		MCDB	37.2	45.1	63.8	76.8	83.5	91.6	94.8	93.8	87.8	73.1	57.7	41.1	
5%		WB	32.7	36.9	50.0	58.0	63.4	67.5	68.1	67.1	61.0	53.8	46.1	35.5	
		MCDB	34.2	40.4	58.9	73.2	82.8	90.9	92.5	92.8	84.0	68.6	52.2	38.0	
10%		WB	30.3	33.4	46.7	55.6	61.6	66.0	66.8	65.5	58.8	51.0	43.1	32.5	
		MCDB	32.1	36.1	54.5	70.2	80.9	89.5	90.9	91.3	79.8	64.4	48.0	34.5	
Mean Daily Temperature Range	5% DB	MDBR	13.7	16.9	20.6	24.2	27.0	27.4	27.2	27.9	28.5	25.1	17.4	13.8	
		MCDBR	11.9	19.0	28.5	30.6	32.2	31.7	30.8	31.9	33.8	32.2	24.6	15.6	
		MCWBR	10.2	14.0	16.0	13.6	13.0	12.1	10.8	12.7	15.2	17.3	16.0	12.6	
	5% WB	MCDWR	11.3	18.6	26.0	27.0	28.0	28.2	28.3	28.8	32.0	29.4	21.6	14.1	
		MCWBR	10.1	14.3	16.0	13.2	13.1	12.4	11.2	12.8	15.4	16.8	14.8	11.7	
Clear-Sky Solar Irradiance	taub	0.348	0.377	0.431	0.471	0.476	0.456	0.441	0.413	0.380	0.388	0.375	0.353		
	taud	2.064	2.025	1.984	1.923	1.955	2.062	2.128	2.187	2.230	2.128	2.115	2.090		
	Ebn at Noon	226	244	248	250	253	259	262	265	264	239	214	209		
	Edh at Noon	32	41	49	57	57	51	48	43	38	36	31	29		
All-Sky Solar Radiation	RadAvg	447	818	1240	1716	2198	2371	2322	2088	1637	1026	576	377		
	RadStd	96	101	98	98	125	120	116	105	77	58	63	58	58	